

# Positional Information Causally Constrains Background Shortcut Reliance in Vision Transformer Models

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## Abstract

*Vision Transformers (ViTs) match or exceed the strongest architectures for object recognition, but, like Convolutional Neural Networks (CNNs), they can exploit image backgrounds rather than foreground, focused objects. Whether a ViT's positional encoding causally governs this reliance has not been thoroughly tested in the literature, and it is also unclear whether rotary encodings (RoPE) constrain such shortcuts more effectively than absolute encodings (APE). In this work, we probe this assumption by scaling the positional signal of eight identically trained ViT checkpoints and measuring their resulting changes in background reliance on ImageNet-9. We find that attenuating the positional signal monotonically raises background reliance in every model (e.g. 5.1  $\rightarrow$  12.8 pp), that the effect is invariant to the encoding scheme, and that the effect is specific to spatial information loss. These findings extend to the Waterbirds dataset [8]. We investigate several mechanisms and find little evidence that any single mechanism carries the effect.*

## 1. Introduction

### 1.1. The Robustness Problem

Robust object recognition can be undermined when a model relies on spurious background cues rather than foreground objects. Modern recognition models, including Vision Transformers (ViTs), are known to exploit such background shortcuts; for instance, by predicting “fish” from surrounding water rather than from the fish itself [4, 12]. This dependence may be unclear in standard test distributions but arises when backgrounds and labels are decoupled and shifted [8]. Understanding what constrains a model from relying on these shortcuts is therefore important for understanding and improving robustness.

### 1.2. The Role of Positional Information

A ViT processes an image as an unordered set of patches, depending on positional information to recover the spatial

layout. We study two families of positional encodings: absolute positional encodings (APE), which add position-dependent embeddings, and rotary positional encodings (RoPE), which introduce position-dependent rotations into attention. [5]. Because RoPE encodes relative geometry, it is theorized to support more robust spatial reasoning, thus being better at resisting background shortcuts than APE. Whether this distinction affects background reliance remains unclear. More fundamentally, it is unknown whether the positional signal causally constrains the shortcut in any capacity.

### 1.3. Research Question & Approach

Our contributions are as follows.

- We show that attenuating the functional positional signal monotonically increases background reliance across all eight checkpoints (e.g., 5.1  $\rightarrow$  12.8 pp), isolated by a within-model dissociation between functional and vestigial signals.
- Across the matched checkpoints studied here, positional encoding variants are pairwise equivalent in background reliance within a 2 pp margin, providing a powered null for a practically meaningful encoding effect.
- We show the effect is specific to spatial-informational loss rather than generic degradation, the effect is replicated on a second spurious benchmark (Waterbirds) [8], and four candidate mechanisms fail to localize the effect to any single tested channel.

The code, evaluation scripts, and experiment configurations are available at <https://github.com/sks17/posvit>.

Figure 1 summarizes our setup and the main result.

## 2. Related Work

**Background bias and spurious correlations.** Prior work establishes that object-recognition models often exploit background elements. Contextual bias research precedes

deep learning [9, 13], and Geirhos et al. [4] later situated such behavior within the broader framework of shortcut learning. Xiao et al. [12] provided a controlled benchmark by constructing ImageNet-9 / the Backgrounds Challenge, quantifying the reliance differences between backgrounds and foregrounds. We adopt this BG-Gap metric. Waterbirds [8] provides a second benchmark with spurious correlations between bird labels and land or water backgrounds, and methods such as DFR [7] and background masking [1] have been used to reduce reliance on spurious features. These works characterize or remove the shortcut. We build upon these findings that characterize or remove background shortcuts by intervening to search for their causative internal factors.

**Positional encodings in Vision Transformers.** Because self-attention does not inherently represent the spatial arrangement of image patches, ViTs require an explicit positional signal to encode their patches [3]. Absolute positional encodings (APE) add learned per-position offsets while relative and rotary encodings store geometry through rotations of query and key representations. Wu et al. [11] studied relative encodings for vision, and Heo et al. [5] introduced 2D rotary encodings (RoPE-Axial, RoPE-Mixed). Touvron et al. [10] provide the DeiT-III checkpoints we use.

**Causal and mechanistic analysis.** Within ViTs, interpretability studies have identified register / artifact tokens (high-norm tokens that process global information rather than patch-level findings) [2]. More broadly, prior work has cautioned against treating attention weights alone as explanations for model predictions [6]. [6]. We manipulate the positional signal while holding model weights fixed and measure the resulting change in background reliance. We then test several candidate mechanisms to determine whether the causal effect can be localized to a particular internal channel.

### 3. Methodology

Broadly, we aim to test whether positional signal, when isolated, controls background-shortcut reliance. We describe the models (Sec. 3.1), the benchmark and metric (Sec. 3.2), the intervention (Sec. 3.3) and its control (Sec. 3.4), the accuracy-matched specificity controls (Sec. 3.5), the interpretability guard (Sec. 3.6), our statistical approach (Sec. 3.7), the cross-dataset generality test (Sec. 3.8), and the mechanism probes (Sec. 3.9). Figure 1 summarizes the general pipeline.

#### 3.1. Models

We study eight DeiT-III checkpoints from the public RoPE-ViT release [5, 10]: five ViT-S/16 and three ViT-B/16 models spanning APE, RoPE-Axial, RoPE-Mixed, and hybrid variants. Within each model scale, checkpoints use the same training recipe and differ primarily in their positional

| Encoding         | Scales | Clean Acc. (%) |
|------------------|--------|----------------|
| APE              | S, B   | 97.1           |
| RoPE-Axial       | S      | 97.3           |
| RoPE-Axial + APE | S      | 97.1           |
| RoPE-Mixed       | S, B   | 97.2           |
| RoPE-Mixed + APE | S, B   | 97.4           |

Table 1. The eight matched checkpoints (five ViT-S/16, three ViT-B/16), differing only in positional encoding. Accuracy is clean ImageNet-9 top-1 for the ViT-S/16 models; the three ViT-B/16 models score  $\sim 98\%$ .

encoding. Their positional encodings are APE, RoPE-Axial, RoPE-Axial + APE, RoPE-Mixed, and RoPE-Mixed + APE with five ViT-S/16 and three ViT-B/16. RoPE-Axial applies rotary embeddings independently along vertical and horizontal axes. RoPE-Axial + APE combines axial rotary encoding with absolute positional encoding. RoPE-Mixed uses a learned mixture of horizontal and vertical components to provide a more flexible representation than RoPE-Axial. RoPE-Mixed + APE augments mixed rotary positional encoding with absolute positional encoding, allowing for the most flexible representation. We use the released checkpoints and train no models, removing training factors as confounds. Importantly, this introduces the limitation (Sec. 5.4) of a single seed per encoding. All models are loaded by explicit checkpoint restoration. Table 1 lists the models and their clean ImageNet-9 accuracies.

#### 3.2. Benchmark & Metrics

We measure background reliance on ImageNet-9 / the Backgrounds Challenge [12], a nine-class benchmark with foreground-paired variants—a fixed foreground object composed onto different backgrounds, separated by class. We use two main variants: MIXED-SAME places the object on a randomly chosen same-class background and MIXED-RAND on a random-class background. We quantify reliance as the background gap,

$$\text{BG-Gap} = \text{Acc}(\text{MIXED-SAME}) - \text{Acc}(\text{MIXED-RAND}), \quad (1)$$

where larger values indicate greater dependence on the background. Because each foreground appears in both variants, Eq. (1) is a *paired* contrast, which enables the paired statistics of Sec. 3.7. Predictions from the 1000-way ImageNet head are mapped into the nine main classes.

#### 3.3. The Causal Intervention

We use a within-checkpoint intervention on the *frozen* positional parameters. For a model and scale factor  $\beta$ , we (i) snapshot the positional parameters, (ii) replace them with a  $\beta$ -scaled copy, (iii) run inference over all variants, and (iv) restore the original parameters. For APE, we scale the



Figure 1. **Within-checkpoint positional intervention (no retraining).** A frozen, identically-trained ViT is evaluated on a background-swapped pair (MIXED-SAME/MIXED-RAND); only the positional signal is scaled by  $\beta$ , and we read off BG-Gap =  $\text{Acc}(\text{MIXED-SAME}) - \text{Acc}(\text{MIXED-RAND})$ . Because the weights are never updated, any change in BG-Gap is caused by the positional signal.

additive positional embedding  $\mathbf{P}$ , and for RoPE we scale the rotation angle  $\theta_p$  applied to queries and keys at position  $p$ :

$$\mathbf{P}(\beta) = \beta \mathbf{P} \quad (\text{APE}), \quad \theta_p(\beta) = \beta \theta_p \quad (\text{RoPE}). \quad (2)$$

Thus  $\beta=1$  does not modify the model,  $\beta=0$  removes all positional information, and  $\beta>1$  amplifies positional information; we sweep  $\beta \in [0, 1]$  for attenuation (and  $[1, 2]$  for amplification) in steps of 0.1. Importantly, the weights are never updated and only  $\beta$  changes between runs, so any changes in BG-Gap are attributable to effects stemming from positional signal. Figure 1 depicts the general pipeline. To test whether the result depends on scalar attenuation itself, we repeat the intervention using structurally different reductions of positional information: spatial low-pass filtering for APE and rotary-frequency truncation for RoPE.

### 3.4. Functional vs. Vestigial Signal

Hybrid models carry both an APE and a RoPE signal but use only one. We define the *functional* signal as the one the architecture relies on and the *vestigial* signal as the unused remainder. Importantly, the unused remainder remains present in the forward computation and can, in principle, influence predictions. We apply the intervention of Sec. 3.3 to each separately. This yields a control—if it is the used positional information that constrains the shortcut, only the functional knob should move BG-Gap. Because the contrast is internal to a single frozen checkpoint, it is not affected by the single-seed limitation of lacking cross-encoding comparisons.

### 3.5. Accuracy-Matched Specificity Controls

To establish that the effect is specific to spatial information rather than destructive interference on the model, we compare positional attenuation against four non-positional corruptions, each swept to the same accuracy as the positional intervention. Patch-wise noise and channel dropout are used to decrease feature fidelity without removing spatial layout. Two spatial controls remove spatial evidence

by other means (token dropout, attention-temperature flattening). For each control we report the excess BG-Gap of positional attenuation at matched accuracy. This is obtained by interpolating each curve to a common accuracy and differencing. A positive excess over a control indicates the positional effect is not explained by that control’s failure mode, thereby distinguishing the spatial information-relevant variables.

### 3.6. Interpretability Guard

We note that BG-Gap is interpretable only while the model maintains a certain baseline of performance. Once an intervention collapses towards chance, the gap becomes non-monotonic and no longer measures background reliance. We therefore impose a floor and interpret BG-Gap only where ORIGINAL-image accuracy exceeds 0.80 on ImageNet-9 (0.70 on Waterbirds, Sec. 3.8). Points beyond the floor are shown for completeness with open markers but are excluded from slopes and tests. 0.8 and 0.7 were empirically chosen as conservative lower bounds based on non-attenuated model performance on their respective datasets.

### 3.7. Statistics

All evaluations are logged per image, allowing for paired inference over the foreground-paired structure. We report 95% confidence intervals from a paired bootstrap that re-samples foreground identities (the clustering unit), McNemar tests for paired accuracy differences, and, to compare the encodings, two one-sided tests (TOST) of equivalence at a 2 pp margin. Monotone dose-response is summarized by Spearman  $\rho$  with a bootstrap slope CI. Cross-encoding multiplicity is controlled by Benjamini–Hochberg FDR across the eight models. A power analysis run across the models gives a minimum detectable effect of 1.6 pp at 80% power, which justifies our equivalence result as a powered null rather than a lack of evidence.

### 3.8. Generality Protocol: Waterbirds

To test generality on a mechanistically different spurious cue, we repeat the process on Waterbirds [8], where bird

species are spuriously correlated with land and water backgrounds. We train a single linear ERM head on frozen ViT-S features under the natural 95%-spurious distribution—so the head *learns* the shortcut. Importantly, we hold it fixed across the  $\beta$  sweep, as refitting at every value of  $\beta$  would erase the effect we intend to study. We measure the spurious gap  $\text{Acc}(\text{congruent}) - \text{Acc}(\text{incongruent})$ , reporting its accuracy-guarded change under positional removal.

### 3.9. Candidate-Mechanism Probes

Finally, we empirically probe four candidate mechanisms for the effect. We test whether (i) high-norm register tokens [2] in background patches predict per-image reliance, (ii) a causal attention-temperature sweep can reproduce the positional effect, (iii) a linear probe can recover the background class from frozen features as the position is attenuated, and (iv) a specific high-frequency rotary band shows the effect at a matched accuracy. These probes test whether the observed positional effect can be explained by a small number of plausible internal mechanisms; their results are reported in Sec. 4.5.

## 4. Experiments and Results

### 4.1. Position Causally Constrains the Shortcut

We first test if attenuating the positional signal raises background reliance in the ViT checkpoints. Sweeping  $\beta$  from 1 to 0 on each frozen model (Fig. 2, left), BG-Gap increases monotonically in all 8/8 checkpoints, with Spearman  $\rho \geq 0.98$  and bootstrap slope CIs excluding zero. All of these results pass BH-FDR at  $p < 10^{-4}$ . The effect holds at both model scales and for all encoding family permutations. Degrading the vestigial signal in hybrids does nothing (Fig. 2, right): its slope brackets zero, and the functional-minus-vestigial difference excludes zero, isolating the effect to the positional information the model uses. We detect no reliable reduction in BG-Gap from amplifying position ( $\beta > 1$ ); all estimated changes have confidence intervals spanning zero.

### 4.2. Background Reliance Is Equivalent Across the Tested Encodings

We test if the encoding scheme changes results. Across the five encodings, intact BG-Gap spans only 4.2–5.1 pp (Fig. 3); under a paired test no pair differs (largest, APE–Axial, +0.94 pp, CI  $[-0.22, +2.10]$ ), and every pair is TOST-equivalent at a 2 pp margin (6/10 at 1.5 pp). With a minimum detectable effect of 1.6 pp, this provides evidence of practical equivalence among the tested checkpoints within our pre-specified 2 pp margin. APE and RoPE are shown to concentrate positional energy in the same low-frequency band through a 2D-Fourier analysis, helping explain why the models are interchangeable.

| Control (what it degrades)     | $\Delta@acc$ (pp) | $\Delta@conf$ (pp) |
|--------------------------------|-------------------|--------------------|
| Patch-noise (feature fidelity) | <b>+3.77</b>      | <b>+4.32</b>       |
| Attn-temperature (routing)     | +1.25             | +2.76              |
| Token-dropout (spatial)        | −0.99             | +0.86              |

Table 2. Excess BG-Gap of positional attenuation over each control at matched accuracy / matched confidence (mean over 8 models). Positive = position-specific. **Bold** = largest excess.

### 4.3. The Effect Is Specific to Spatial Information

We test if any accuracy-matched degradation would have a similar effect. Matching each control to the positional intervention’s clean accuracy (Fig. 4), positional attenuation induces +3.8 pp excess BG-Gap over patch-noise and +1.3 pp over attention-temperature, with CIs excluding zero (Tab. 2). The two spatial controls are comparable, seemingly serving as evidence against the thesis. Their comparability, however, because they too remove spatial information, reinforces the thesis.

### 4.4. Generalization, Robustness, and What Governs the Shortcut

We then test external validity and what governs the shortcut. On Waterbirds (Fig. 5a), removing positional information widens the spurious gap in 5/5 encodings under the outlined accuracy guard, with a gap spread of 1.9 pp that remains encoding-agnostic. Background reliance is strongly associated with foreground object size (Fig. 5b): BG-Gap falls from  $\sim 10$  pp for the smallest foregrounds to  $\sim 1$  pp for the largest, identically across encodings. This re-confirms [12] and, further, the causal result: it is robust to the specific attenuation (low-pass, frequency-truncation) and concentrates in later layers.

### 4.5. Investigating Candidate Mechanisms

Finally we ask what carries the observed effect. We investigated four plausible mechanisms and found little evidence that any single channel does (Tab. 3): while per-image attention entropy can predict background-reliant images (AUC 0.70 [0.67, 0.72]; BG-Gap rising  $0.4 \rightarrow 11.5$  pp across entropy quartiles), register/artifact tokens are *anti*-predictive of per-image reliance (AUC 0.35); a causal attention-temperature sweep is null or reversed; background decodability *falls* rather than rises as the gap grows; and a high-frequency rotary band shows  $\approx 0$  pp excess at matched accuracy. However, the association weakens substantially on Waterbirds (AUC  $\approx 0.54$ ), arguing against attention entropy as a dataset-general mechanism.



## Causal Dose-Response of the Positional Signal

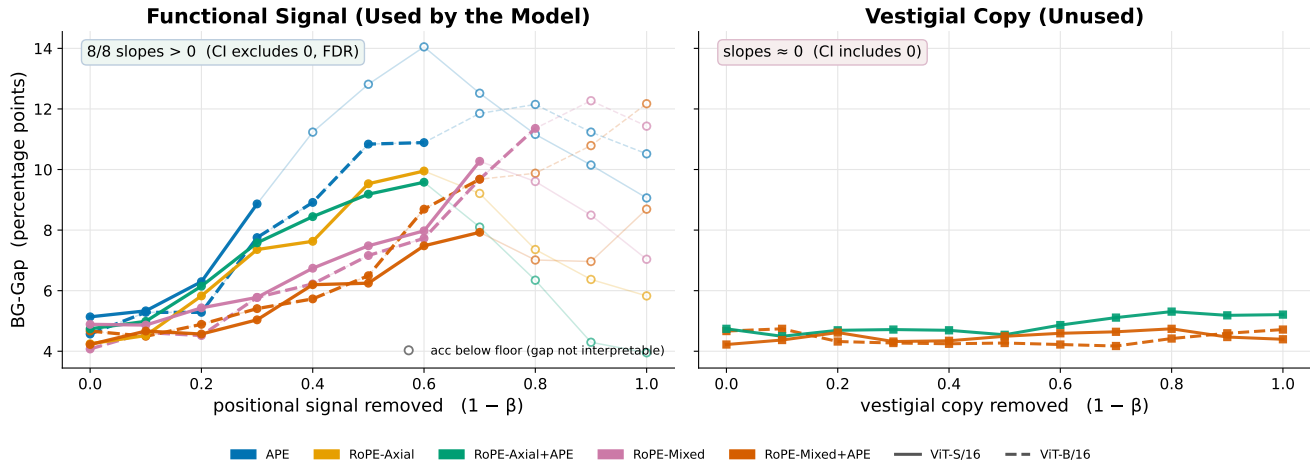


Figure 2. **Attenuating the functional positional signal causally raises background reliance (8/8), while the vestigial signal is inert.** BG-Gap vs. positional removal ( $1 - \beta$ ); solid = above the interpretability floor, open = below. *Left*: functional signal. *Right*: vestigial copy.

| Candidate mechanism        | How tested               | Result                                 |
|----------------------------|--------------------------|----------------------------------------|
| Artifact/“register” tokens | per-image AUC            | 0.35 (anti-predictive)                 |
| Attention geometry/routing | attn-temp sweep          | null / reversed                        |
| Background decodability    | linear probe vs. $\beta$ | falls as gap rises                     |
| High-frequency rotary band | excess @ matched acc.    | $\approx 0$ pp (deny)                  |
| Attention entropy (spread) | per-image AUC            | 0.70 [0.67,0.72] ( <i>diagnostic</i> ) |

Table 3. Four candidate mechanisms are unsupported; one correlational diagnostic survives. The effect resists single-channel localization.

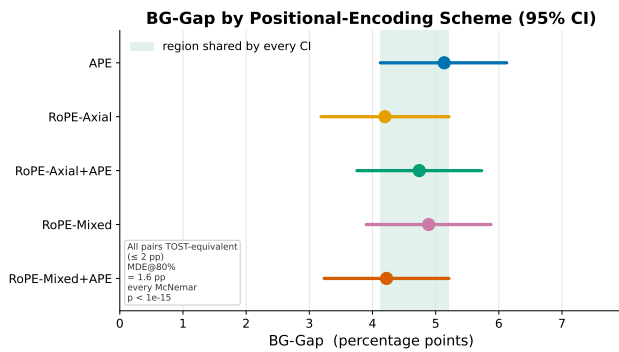


Figure 3. **Background reliance is equivalent across the tested positional-encoding variants within a 2 pp margin.** Intact BG-Gap by encoding with 95% CIs; all pairs are TOST-equivalent at  $\leq 2$  pp (MDE 1.6 pp): a powered null.

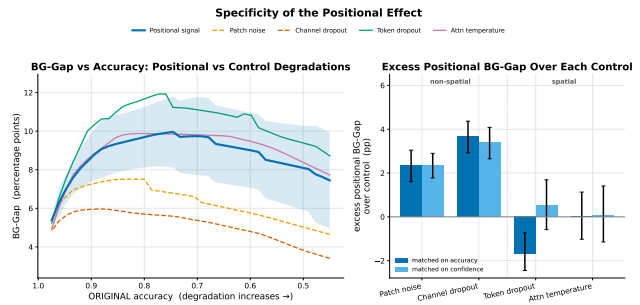


Figure 4. **The increase in background reliance tracks spatial information.** Excess BG-Gap of positional attenuation over accuracy-matched controls; positive vs. non-spatial controls, on par with spatial ones.

## 5. Discussion

### 5.1. What It Means

The results show that the functional positional signal causally constrains a Vision Transformer’s reliance on

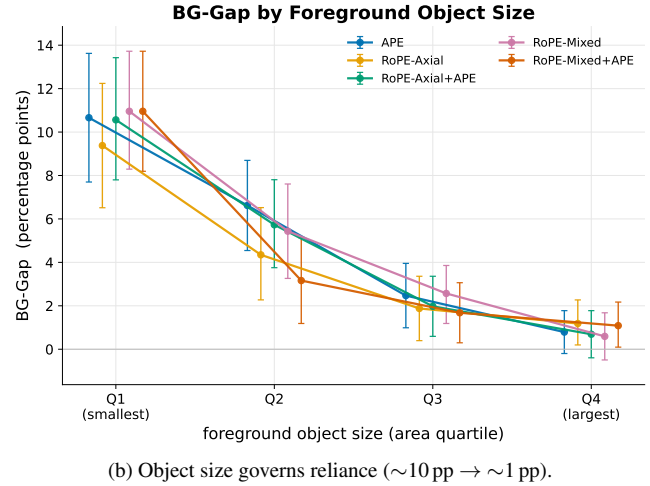
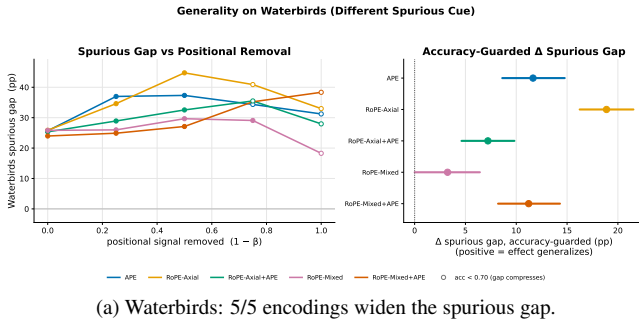


Figure 5. The effect generalizes and is governed by object size.

the background shortcut. This conclusion is supported across encoding variants, degradation controls, and a second benchmark. Attenuating the functional signal increases background reliance in every checkpoint, while the vestigial signal is inert. The effect is also stable across APE, RoPE, and hybrid variants within the tested equivalence margin. Accuracy-matched controls further link the increase in BG-Gap to disrupted spatial information rather than generic performance loss. This finding is not an artifact of one benchmark, as the results are replicated on Waterbirds, and its magnitudes vary systemically with foreground object size.

## 5.2. Mechanisms Remain Unresolved

A natural next question is what carries the constraint. We tested artifact tokens, attention geometry, background decodability, and high-frequency rotary components, but none of these candidates accounts for the observed effect. As such, the constraint, empirically, seems to resist single-channel localization, instead having a more distributed effect.

## 5.3. Necessity, Not Sufficiency

Our evidence remains unbalanced. Attenuation consistently increases background reliance, whereas amplification produces no detectable reduction; all amplification confidence intervals include zero. Therefore, we describe positional information as constraining the shortcut rather than further safeguarding from it.

## 5.4. Limitations

Each encoding is represented by a single checkpoint and training seed, so between-encoding comparisons carry the theoretical idiosyncrasies of each checkpoint. As our causal claims are within checkpoints to defuse this, some of the ad-

verse effects of this are mitigated, but controlled multi-seed training remains. Both ImageNet-9 and Waterbirds study background-based spurious correlations, so broader generalization remains untested. The attention-entropy diagnostic is correlational and may reflect other factors such as example difficulty.

## 6. Conclusion

### 6.1. Summary

In this work, we tested whether positional information causally constrains a Vision Transformer’s reliance on the background shortcut. We did this by scaling the positional signal of eight frozen models that use different positional encoding schemes and measuring the changes in background reliance. Attenuation monotonically increases background reliance in all eight checkpoints, while the tested APE and RoPE variants remain equivalent within a 2 pp margin. This pattern reflects loss of spatial information specifically and generalizes to a second spurious benchmark. The asymmetry between attenuation and amplification suggests, rather than a signal whose simple amplification yields further robustness, positional information acts as a baseline constraint on shortcut reliance.

### 6.2. Future Work

There are several potential directions following this work. Multi-seed training could establish whether the cross-encoding equivalence survives checkpoint variation and whether objectives such as positional dropout strengthen the observed constraint on background reliance. Distributed causal tracing could test whether the effect is localized across multiple layers or components. If the observed constraint can become stronger, it could serve as a

cheap, retraining-free adjustment for robustness.

In application, testing whether the principle that a model's sense of place constrains spurious reliance holds for video models, medical-imaging models, satellite-imagery models, and larger foundation models could help work towards forming a safeguard against these behaviors. Through these sensitive fields, the measured benefit of limiting background reliance could be evidenced.

Our analysis uses only released checkpoints, adding no generative capability and instead primarily being diagnostic.

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